Business Intelligence – Exercise Sheet: CART Decision Trees for Wirtschaftsinformatik

Bachelor in Wirtschaftsinformatik, 6th Semester

15 May 2026

Exercises on CART Classification and Regression Trees

Exercises 1 to 10 concern manual CART classification trees for business-information systems. Exercises 11 to 15 concern CART regression trees for KPI prediction. Exercises 16 to 20 concern pruning, cost-complexity, model evaluation, and interpretation for BI use cases.

Use exact expressions whenever convenient and round numerical answers to three decimals when needed. In classification exercises, use the CART convention of binary splits. For a node t, let $p_k(t)$ be the class proportion of class k. The Gini impurity is

$$G(t) = 1 - \sum_{k=1}^{K} p_k(t)^2.$$

For a split of node t into left child t_L and right child t_R, use the weighted post-split impurity

$$G_{\text{split}} = \frac{n_L}{n} G(t_L) + \frac{n_R}{n} G(t_R),$$

and the impurity reduction

$$\Delta G = G(t) - G_{\text{split}}.$$

For regression trees, use squared-error impurity. For a node t with target values $y_1, \ldots, y_n$, the prediction is the node mean

$$\hat{y}(t) = \bar{y}_t,$$

and the node sum of squared errors is

$$SSE(t) = \sum_{i \in t} (y_i - \bar{y}_t)^2.$$

For a regression split, use

$$SSE_{\text{split}} = SSE(t_L) + SSE(t_R), \qquad \Delta SSE = SSE(t) - SSE_{\text{split}}.$$

For cost-complexity pruning, use

$$R_\alpha(T) = R(T) + \alpha |T|,$$

where $R(T)$ is the empirical loss of the tree, $|T|$ is the number of terminal leaves, and $\alpha \geq 0$ is the complexity penalty. For classification evaluation, use

$$\text{accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \qquad \text{precision} = \frac{TP}{TP + FP}, \qquad \text{recall} = \frac{TP}{TP + FN},$$

$$F_1 = \frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}.$$

1. **Ex 1 (20 Points):** A BI team wants to predict whether a customer will churn from a subscription platform. The current root node contains 10 customers:

$$6 \text{ churn}, \qquad 4 \text{ no churn}.$$

A candidate split by the binary feature

$$\text{SupportTicketsHigh} \in \{\text{yes}, \text{no}\}$$

produces:

$$\begin{aligned} t_L : 5 \text{ customers} &= 4 \text{ churn}, 1 \text{ no churn}, \\ t_R : 5 \text{ customers} &= 2 \text{ churn}, 3 \text{ no churn}. \end{aligned}$$

   (a) Compute the Gini impurity of the root node. (5 Points)

   (b) Compute the Gini impurity of t_L and t_R. (6 Points)

   (c) Compute the weighted post-split Gini impurity. (5 Points)

   (d) Compute the impurity reduction and explain whether the split is useful. (4 Points)

2. **Ex 2 (20 Points):** An ERP vendor wants to classify whether a lead converts to a paying customer. The root node contains the following 12 leads:

$$7 \text{ converted}, \qquad 5 \text{ not converted}.$$

Two candidate CART splits are tested.

Split A: DemoRequested = yes

$$t_L : 6 \text{ leads} = 5 \text{ converted}, 1 \text{ not converted},$$

$$t_R : 6 \text{ leads} = 2 \text{ converted}, 4 \text{ not converted}.$$

Split B: CompanySizeLarge = yes

$$t_L : 5 \text{ leads} = 4 \text{ converted}, 1 \text{ not converted},$$

$$t_R : 7 \text{ leads} = 3 \text{ converted}, 4 \text{ not converted}.$$

   (a) Compute the root Gini impurity. (4 Points)

   (b) Compute the weighted post-split Gini impurity for Split A. (6 Points)

   (c) Compute the weighted post-split Gini impurity for Split B. (6 Points)

   (d) Which split does CART select? Justify using ΔG. (4 Points)

3. **Ex 3 (20 Points):** A Wirtschaftsinformatik project team classifies whether an IT ticket is escalated. The data are:

Ticket	Waiting time x in hours	Priority	Escalated?
1	1	low	no
2	2	low	no
3	3	medium	no
4	4	medium	yes
5	5	high	yes
6	6	high	yes

Consider threshold splits of the numerical variable x.

(a) List the possible CART thresholds between consecutive sorted values of x. (4 Points)

(b) Compute the weighted Gini impurity for the split $x \leq 3.5$. (6 Points)

(c) Compute the weighted Gini impurity for the split $x \leq 4.5$. (6 Points)

(d) Which of these two splits is better? Interpret the selected threshold in IT-service terms. (4 Points)

4. **Ex 4 (20 Points):** A CRM system classifies customers into

$$\text{high value} \qquad \text{or} \qquad \text{standard value.}$$

At a candidate node there are 8 customers:

$$3 \text{ high value}, \qquad 5 \text{ standard value.}$$

A split by

$$\text{UsesMobileApp} \in \{\text{yes}, \text{no}\}$$

produces:

$$\begin{aligned} t_L : 3 \text{ customers } &= 3 \text{ high value}, 0 \text{ standard value}, \\ t_R : 5 \text{ customers } &= 0 \text{ high value}, 5 \text{ standard value}. \end{aligned}$$

(a) Compute the Gini impurity of the parent node. (5 Points)

(b) Compute the Gini impurity of both child nodes. (5 Points)

(c) Compute the impurity reduction. (5 Points)

(d) Explain why this split is called a pure split and why it is attractive for a decision tree. (5 Points)

5. **Ex 5 (20 Points):** A digital-sales dashboard stores the following observations:

Customer	Newsletter	App user	Orders last month	Buy premium?
1	yes	yes	4	yes
2	yes	no	3	yes
3	yes	yes	2	yes
4	no	yes	2	no
5	no	no	1	no
6	no	no	0	no
7	yes	no	1	no
8	no	yes	3	yes

(a) Compute the Gini impurity of the root node for the target variable Buy premium?. (4 Points)

(b) Compute the weighted Gini impurity for the split Newsletter = yes. (6 Points)

(c) Compute the weighted Gini impurity for the split App user = yes. (6 Points)

(d) Which categorical split is better at the root? Explain in BI language. (4 Points)

6. **Ex 6 (20 Points):** A bank uses a CART model to classify whether a customer accepts a digital-credit offer. The current node has three classes:

accept, reject, undecided.

At the parent node:

4 accept, 3 reject, 3 undecided.

A candidate split gives:

$$t_L : 5 \text{ customers } = 4 \text{ accept}, 1 \text{ reject}, 0 \text{ undecided},$$

$$t_R : 5 \text{ customers } = 0 \text{ accept}, 2 \text{ reject}, 3 \text{ undecided}.$$

(a) Compute the multiclass Gini impurity of the parent node. (5 Points)

(b) Compute the multiclass Gini impurity of t_L. (4 Points)

(c) Compute the multiclass Gini impurity of t_R. (4 Points)

(d) Compute the weighted post-split impurity and the impurity reduction. (5 Points)

(e) Assign a predicted class to each child node using majority vote. (2 Points)

7. **Ex 7 (20 Points):** A BI analyst builds the first level of a CART classification tree for predicting whether an online order is returned. The root data are:

Order	Delivery delay x in days	Basket value v	Returned?
1	0	40	no
2	1	55	no
3	2	80	no
4	3	95	yes
5	4	120	yes
6	5	140	yes
7	6	160	yes
8	7	200	yes

(a) Compute the root Gini impurity. (4 Points)

(b) Compute the weighted Gini impurity for the split $x \leq 2.5$. (5 Points)

(c) Compute the weighted Gini impurity for the split $x \leq 3.5$. (5 Points)

(d) Compute the weighted Gini impurity for the split $v \leq 100$. (4 Points)

(e) Which split is selected by CART? (2 Points)

8. **Ex 8 (20 Points):** A decision tree node contains 14 observations for the target

$$\text{Invoice paid late?}$$

The parent node contains:

$$8 \text{ late}, \qquad 6 \text{ on time}.$$

A candidate split by

$$\text{CustomerSegment} \in \{\text{A}, \text{B}, \text{C}\}$$

must be converted into binary CART splits. The segment counts are:

$$A : 5 \text{ observations} = 4 \text{ late}, 1 \text{ on time},$$

$$B : 4 \text{ observations} = 1 \text{ late}, 3 \text{ on time},$$

$$C : 5 \text{ observations} = 3 \text{ late}, 2 \text{ on time}.$$

Consider the binary splits:

$$\{A\} \mid \{B, C\}, \qquad \{B\} \mid \{A, C\}, \qquad \{C\} \mid \{A, B\}.$$

(a) Compute the root Gini impurity. (4 Points)

(b) Compute the weighted Gini impurity for $\{A\} \mid \{B, C\}$. (5 Points)

(c) Compute the weighted Gini impurity for $\{B\} \mid \{A, C\}$. (5 Points)

(d) Compute the weighted Gini impurity for $\{C\} \mid \{A, B\}$. (4 Points)

(e) Which binary categorical split does CART select? (2 Points)

9. **Ex 9 (20 Points):** A CART model for detecting potential fraud in expense claims has reached the following terminal leaves:

Leaf	Fraud cases	Non-fraud cases
1	5	1
2	1	7
3	2	2

   (a) Compute the predicted class of each leaf using majority vote. (4 Points)

   (b) Compute the training misclassification count of each leaf. (6 Points)

   (c) Compute the total training misclassification count of the tree. (4 Points)

   (d) Compute the total training accuracy of the tree. (4 Points)

   (e) Explain why Leaf 3 may be considered operationally ambiguous. (2 Points)

10. **Ex 10 (20 Points):** A company uses a CART classifier for churn prediction. A terminal leaf contains 20 customers:

$$15 \text{ churn}, \qquad 5 \text{ no churn}.$$

   (a) Compute the class probability estimates produced by the leaf. (4 Points)

   (b) State the predicted class using the default majority-vote rule. (4 Points)

   (c) Suppose the business cost of missing a churner is four times larger than the cost of incorrectly contacting a non-churner. Explain why the default majority-vote threshold may not be optimal. (6 Points)

   (d) If the retention team contacts all customers with predicted churn probability at least 0.6, determine whether customers in this leaf are contacted. (3 Points)

   (e) If the threshold is changed to 0.8, determine whether customers in this leaf are contacted. (3 Points)

11. **Ex 11 (20 Points):** A BI team builds a regression tree to predict monthly revenue per customer in EUR. At a node, the target values are

$$y = (100, 120, 130, 150, 300, 320).$$

A candidate split by

$$\text{PremiumAccount} = \text{yes}$$

produces:

$$t_L : y = (300, 320), \qquad t_R : y = (100, 120, 130, 150).$$

   (a) Compute the parent-node prediction and $SSE(t)$. (6 Points)

(b) Compute the prediction and SSE for t_L. (5 Points)

(c) Compute the prediction and SSE for t_R. (5 Points)

(d) Compute ΔSSE and interpret the split in revenue-management terms. (4 Points)

12. **Ex 12 (20 Points):** A regression tree predicts monthly service cost from the number of open support tickets x. The data are:

Customer	Tickets x	Service cost y in EUR
1	0	40
2	1	45
3	2	55
4	3	80
5	4	90
6	5	95

Consider the split $x \leq 2.5$.

(a) Compute the parent mean and parent SSE. (6 Points)

(b) Compute the left-child mean and SSE. (5 Points)

(c) Compute the right-child mean and SSE. (5 Points)

(d) Compute the SSE reduction and explain why the threshold is meaningful for service controlling. (4 Points)

13. **Ex 13 (20 Points):** A regression tree predicts the number of monthly app logins from customer segment behavior. The target values at a node are:

$$y = (5, 6, 7, 20, 22, 24, 25).$$

Two candidate splits are tested.

Split A:

$$t_L : y = (5, 6, 7), \qquad t_R : y = (20, 22, 24, 25).$$

Split B:

$$t_L : y = (5, 6, 7, 20), \qquad t_R : y = (22, 24, 25).$$

(a) Compute the parent-node mean and SSE. (5 Points)

(b) Compute SSE_{split} for Split A. (5 Points)

(c) Compute SSE_{split} for Split B. (5 Points)

(d) Which split does CART select? Justify with ΔSSE. (5 Points)

14. **Ex 14 (20 Points):** A BI dashboard uses a regression tree to predict order volume. The following terminal leaves are obtained:

Leaf	Number of observations	Sum of actual y	Within-leaf SSE
1	4	80	30
2	5	200	50
3	3	180	24

(a) Compute the prediction assigned to each terminal leaf. (6 Points)

(b) Compute the total training SSE of the regression tree. (4 Points)

(c) A new observation falls into Leaf 2. What is its predicted order volume? (4 Points)

(d) Explain why a regression tree prediction is piecewise constant. (4 Points)

(e) Briefly state one advantage and one disadvantage of piecewise-constant BI predictions. (2 Points)

15. **Ex 15 (20 Points):** A regression tree predicts monthly data-processing cost. At a node, the target values are:

$$y = (10, 12, 13, 15, 40, 42, 45, 48).$$

A numerical split by

$$\text{DataVolume} \leq 1000$$

produces:

$$t_L : y = (10, 12, 13, 15), \qquad t_R : y = (40, 42, 45, 48).$$

(a) Compute the parent-node mean and $SSE(t)$. (6 Points)

(b) Compute the two child-node means and their SSE values. (6 Points)

(c) Compute ΔSSE. (4 Points)

(d) Explain why this split may be a strong candidate even if the data set is small. (4 Points)

16. **Ex 16 (20 Points):** A trained CART classification tree for predicting whether a customer renews a software subscription has three terminal leaves:

Leaf	Number of customers	Renewed	Not renewed
1	30	27	3
2	20	11	9
3	10	2	8

(a) Assign the predicted class to each leaf. (4 Points)

(b) Compute the training misclassification count of each leaf. (6 Points)

(c) Compute the total training error rate. (4 Points)

(d) Which leaf is the least reliable from a decision-support perspective? Justify numerically. (4 Points)

(e) Explain why probability estimates are more informative than only hard class labels in a CRM application. (2 Points)

17. **Ex 17 (20 Points):** A CART classification tree has the following candidate subtrees:

Tree	Training misclassification $R(T)$	Number of leaves $\|T\|$
T_1	10	2
T_2	7	4
T_3	5	8

Use the cost-complexity criterion

$$R_\alpha(T) = R(T) + \alpha|T|.$$

(a) Compute $R_\alpha(T_1)$, $R_\alpha(T_2)$, and $R_\alpha(T_3)$ for $\alpha = 0$. (4 Points)

(b) Compute the same values for $\alpha = 1$. (5 Points)

(c) Compute the same values for $\alpha = 2$. (5 Points)

(d) For each value of α, state which tree is selected. (3 Points)

(e) Explain in BI language why increasing α favors simpler trees. (3 Points)

18. **Ex 18 (20 Points):** A CART churn classifier is evaluated on a test set. The confusion matrix is:

	Predicted churn	Predicted no churn
Actual churn	18	6
Actual no churn	4	32

Treat churn as the positive class.

(a) Identify TP, FP, TN, and FN. (4 Points)

(b) Compute accuracy. (4 Points)

(c) Compute precision and recall. (6 Points)

(d) Compute the F_1-score. (4 Points)

(e) Explain whether the model is more problematic for false positives or false negatives in a retention-management setting. (2 Points)

19. **Ex 19 (20 Points):** A company compares two CART models for detecting risky invoices. The test-set results are:

$$\text{Model A: } TP = 20,\ FP = 10,\ TN = 70,\ FN = 5,$$
$$\text{Model B: } TP = 17,\ FP = 4,\ TN = 76,\ FN = 8.$$

(a) Compute accuracy for both models. (4 Points)

(b) Compute precision for both models. (4 Points)

(c) Compute recall for both models. (4 Points)

(d) Compute the F_1-score for both models. (4 Points)

(e) If missing a risky invoice is very expensive, which model should be preferred? Justify using the metrics. (4 Points)

20. **Ex 20 (20 Points):** A BI project team wants to deploy a CART model in a management dashboard for predicting late delivery risk. The first-level tree rule is:

$$\text{If delivery delay forecast } \leq 2.5 \text{ days, predict low risk;}$$
$$\text{otherwise, check whether supplier rating } \geq 4.$$
$$\text{If supplier rating } \geq 4, \text{ predict medium risk; otherwise predict high risk.}$$

Five new orders are:

Order	Delay forecast in days	Supplier rating
A	1.0	3
B	2.5	5
C	3.0	5
D	4.0	2
E	2.8	4

(a) Apply the tree manually and assign a risk class to each order. (6 Points)

(b) Draw the binary tree structure with its two internal decision nodes and three terminal leaves. (5 Points)

(c) Explain why this model is interpretable for logistics managers. (4 Points)

(d) State two risks of deploying this tree without continuous monitoring. (3 Points)

(e) Name one dashboard KPI that should be monitored to detect model degradation over time. (2 Points)